

respond to commands from a host device. In a USB Type-C ecosystem, a host sends series of discover commands to know about the cable and decide further action on how to set up the ecosystem. Thus, a cable plays a key role in deciding the USB Type-C ecosystem.

Can decide the power a source can provide. During the discover process, a host device receives identity of a Type-C cable that contains details like manufacturer name, type of cable, USB capabilities, V_{BUS} current handling capability and so on. This information about V_{BUS} current capability restricts a power provider its power capability advertisement. This means, even though a power provider is capable of providing 5A to a consumer and the cable in its identity communicates its limits as 1.5A, current capability is restricted to 1.5A.

Can be active or passive. A USB Type-C cable can be categorised as electronically-marked or non-electronically-marked cables, where the latter is dumb and does not contain any electronics inside.

Electronically-marked cables can be further classified as passive cables that are USB Type-C cables that do not support data bus signal-conditioning circuits but can respond to discover commands, and active cables that are Type-C cables that support data bus signal-conditioning circuits along with discover commands.

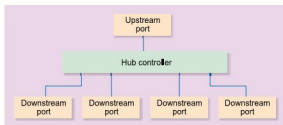


Fig. 5: Legacy USB hub

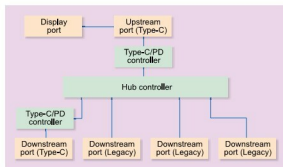


Fig. 6: USB Type-C USB hub solution

This classification of the cables enables bringing more interesting functionalities along with some complications to the end user.

Can enter modes. Type-C connectors allow functionalities like Thunderbolt or HDMI over existing USB signal lines. This is achieved through CC protocol. Such alternate modes are not restricted to a host or a device but can also be extended to a cable. For example, these alternate modes can help an active enable signal conditioning or at times perform some other proprietary action before entering alternate mode between a host and a device.

USB Type-C hub

Another major change the USB Type-C technology brings in is on the USB

hub front, which is now a multi-function device that supports functionalities like Display Port or MHL, along with USB in a USB hub solution.

Let us now explore what a legacy hub solution looks like and how it can be modified to a Type-C solution. Fig. 5 illustrates a simple legacy hub controller with four downstream ports and an upstream port, all supporting USB functionality.

In simple terms, a USB Type-C hub solution is just a tweak of the above legacy solution with Type-C controllers managing ports and signals as illustrated in Fig. 2.

Fig. 6 is a break up of a USB Type hub, which has the following features:

- Three USB-A 3.0
- Mini display port
- Two USB-C

From Fig. 6, it is easy to understand how an existing legacy USB hub design can be tweaked to a USB Type-C hub. It is important to note that in this new hub model a USB Type-C controller manages the signal lines of the USB Type-C connector after successful CC protocol negotiation.

Some key points to remember in a USB Type-C hub solution are:

- Only an upstream port of a hub can support alternate mode, that is, functionality other than USB.
- When operating in display port mode, a USB Type-C hub may not support USB 3.0 functionality and will operate in USB 2.0 mode. **EFY**



For More
Test and Equipment Stories
Visit www.electronicstb2b.com